

4. Project Planning Phase

Introduction

Project Planning is an important phase in the software development life cycle. This phase helps in organizing tasks, allocating resources, identifying risks, and setting project milestones. Proper planning ensures that the project is completed successfully within the given time and resources.

For the AI-Powered Smart Visitor Management System, a structured 5-day development plan was prepared to complete the project efficiently using ServiceNow.

Project Objectives

The main objective of the planning phase is to ensure smooth project execution.

Objectives

- Define project activities.
 - Allocate resources effectively.
 - Identify potential risks.
 - Monitor project progress.
 - Complete the project within the planned schedule.
-

5-Day Development Plan

The project was divided into five development phases.

Day 1 – Application Creation and Table Design

Activities Performed

- Created a new ServiceNow application.
- Defined project scope.
- Created the Visitor Request table.
- Configured application modules.
- Created basic navigation menus.

Deliverables

- Smart Visitor Management Application
- Visitor Request Table
- Application Menu

Outcome

The basic application structure was successfully created.

Day 2 – Form Design and Field Creation

Activities Performed

- Added all required fields.
- Configured form layout.
- Created choice lists.
- Defined field properties.
- Configured auto-generated Visitor ID.

Fields Created

- Visitor ID
- Visitor Name
- Mobile Number
- Email
- Purpose
- Visit Date
- Status
- AI Recommendation

Outcome

The Visitor Request Form was successfully designed.

Day 3 – Client Scripts and Business Rules

Activities Performed

- Created Mobile Number Validation Script.
- Created Visit Date Validation Script.
- Developed AI Recommendation Business Rule.
- Configured Default Status Business Rule.

Business Logic Implemented

Mobile Validation

Checks whether the entered mobile number contains exactly 10 digits.

Date Validation

Prevents users from selecting past dates.

AI Recommendation

Automatically generates recommendations based on visitor purpose.

Outcome

Data validation and automation logic were successfully implemented.

Day 4 – Flow Designer and Notifications

Activities Performed

- Created Visitor Request Flow.
- Configured Flow Trigger.
- Added Send Email Action.
- Configured Email Templates.
- Tested flow execution.

Automation Process

Visitor Request Created ↓ Flow Triggered ↓ Email Generated ↓ Email Sent

Outcome

The notification system was successfully automated.

Day 5 – Reports, Dashboard and Testing

Activities Performed

- Created Visitor Status Report.
- Created Visitor Purpose Report.
- Created Monthly Visitor Report.
- Designed Dashboard.
- Conducted testing.
- Fixed minor issues.

Outcome

The project was fully completed and tested.

Resource Planning

Resource planning ensures that all necessary resources are available during development.

Software Resources

ServiceNow Platform

Used for application development, workflow automation, reporting, and dashboard creation.

Web Browser

Used to access the ServiceNow instance.

Internet Connection

Required to access cloud-based ServiceNow services.

Hardware Resources

Laptop/Desktop

Minimum Specifications:

- Intel Core i3 Processor or above
- 4 GB RAM
- 20 GB Free Storage

Network Connectivity

Stable internet connection for development and testing.

Human Resources

The project can be developed by:

Developer

Responsible for creating forms, scripts, workflows, and dashboards.

Tester

Responsible for testing the application and identifying defects.

End User

Uses the application for visitor registration and tracking.

Risk Analysis

Risk analysis helps identify potential project issues and provides solutions.

Risk 1 – Incorrect Data Entry

Problem

Users may enter invalid information.

Solution

Implement Client Scripts for validation.

Risk 2 – Email Notification Failure

Problem

Email notifications may not be delivered.

Solution

Verify Flow Designer configuration and email settings.

Risk 3 – System Downtime

Problem

ServiceNow instance may be temporarily unavailable.

Solution

Perform testing during stable network conditions.

Risk 4 – Incorrect Status Updates

Problem

Visitor status may not update correctly.

Solution

Use Business Rules and UI Actions for automation.

Risk 5 – Data Duplication

Problem

Duplicate visitor records may be created.

Solution

Use auto-generated Visitor IDs and validation checks.

Timeline Chart

Day	Activity
Day 1	Application Creation and Table Design
Day 2	Form Design and Field Creation
Day 3	Client Scripts and Business Rules
Day 4	Flow Designer and Email Notifications
Day 5	Reports, Dashboard and Testing

Project Milestones

The following milestones were achieved during development.

Milestone 1

Application Created Successfully

Milestone 2

Visitor Request Table Created

Milestone 3

Fields and Forms Configured

Milestone 4

Client Scripts Implemented

Milestone 5

Business Rules Developed

Milestone 6

Flow Designer Activated

Milestone 7

Email Notifications Configured

Milestone 8

Reports Generated

Milestone 9

Dashboard Created

Milestone 10

Testing Completed Successfully

Benefits of Project Planning

- Better project organization.
 - Efficient use of resources.
 - Reduced project risks.
 - Faster development process.
 - Improved project quality.
 - Timely project completion.
-

Conclusion

The Project Planning Phase played a vital role in the successful development of the AI-Powered Smart Visitor Management System. Through proper planning, resource allocation, risk management, and milestone tracking, the project was completed within the planned five-day schedule. This phase ensured smooth execution and contributed significantly to the overall success of the project.